

Discussion of Advances in phylodynamic inference: integrating epidemiological dynamics and genetic data

Ed Ionides

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Similarities among the talks

- ▶ Theory concerning the likelihood function for models of phylodynamic processes.
- ▶ Methodology addressing practical numerical considerations. Usually, this involves tractable approximations.

In the context of hot topics

Large language models. These can help algebraic and numeric calculations in extending the scope of these methods. Usual considerations apply about assessing correctness and maintaining human skill.

Deep learning. Neural networks can be used for nonparametric components of a model, or for building simulation-based estimators.

Ebola. The currently outbreak is already the 2nd largest known.

Bird flu. H5N1 avian influenza is a continuing risk for a human pandemic.

Questions

Scalability. What is the scale (number of sequences, and population size) that your current methodology can handle? How could your approach be scaled further, if that is desired?

Overdispersion. What if processes do not have Poisson dispersion? For Markov compartment models, that implies accommodating simultaneous transition events.

Robustness to misspecification. “All models are wrong, but some models are useful.” What is the clearest evidence about utility despite misspecification for phylodynamic inference? I.e., what are the most impressive existing scientific applications of phylodynamics?

Age of AI. How are AI advances going to affect this area?

Age of data. How are advances in high-throughput data collection going to affect this area?